

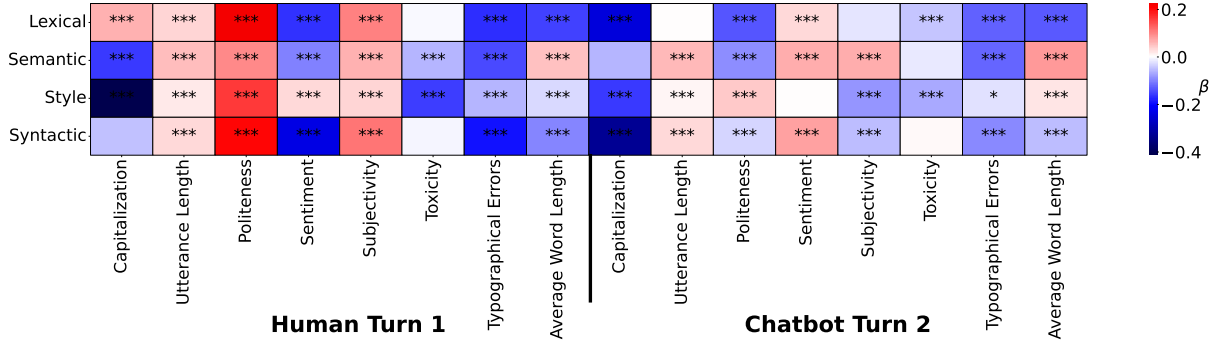


Figure 6: In what contexts do SIMULATORS best predict HUMAN responses? We show the results of four regressions predicting the similarity between SIMULATOR and HUMAN at Turn 3 for different categories (rows), using HUMAN Turn 1 and CHATBOT Turn 2 properties as features (columns). We highlight a subset of the coefficients here, where red and blue colors indicate positive and negative regression coefficients  $\beta$  respectively, and stars in each cell indicate the statistical significance of each  $\beta$  after applying a Bonferroni correction (\*  $p<0.05$ , \*\*  $p<0.01$ , \*\*\*  $p<0.001$ ); Full regression coefficients are given in Table 9 and Table 10. The linguistic properties of HUMAN Turn 1 have stronger effects than those of the CHATBOT in Turn 2, showing that SIMULATORS do correctly accommodate more to the HUMAN style. However, SIMULATORS tend to perform better when the HUMAN’s Turn 1 more closely matches the properties of typical SIMULATOR-generated text (e.g., more polite, fewer typos).

atively smaller amounts of training data. Consequently, safety training may impact these languages’ outputs more significantly than English, leading to a strong prior on toxicity and sentiment that hinders the style match to HUMANS. Additionally, English SIMULATORS demonstrate superior accuracy in modeling syntactic metrics.

The choice of model and prompting strategy affects performance, as shown in Appendix Figure 10b. The differences between models become more pronounced in languages other than English, as shown in Figure 10a. For example, while model differences in English are often minor, the smallest model (MIXTRAL-8x7B) frequently underperforms compared to other models in Chinese and Russian. Moreover, the significant inter-model variation in Chinese and Russian may be due to larger models having more non-English samples in their pre-training data, which enhances their performance in non-English languages. Additionally, the F1 score shows more variation across the models and prompting strategies than the text properties metrics, suggesting that engineering decisions may be more salient for closed-ended conversation-ending tasks than open-ended text generation tasks. Appendix §E details the differences.

Overall, we find that LLMs’ performance as SIMULATORS is largely consistent across English, Chinese, and Russian, although some differences suggest that languages for which SIMULATORS had less training data may have less robust performance across contexts (RQ2).

## 7 When Do LLMs Successfully Simulate?

Experiments from §5 and §6 paint a high-level picture of how the SIMULATORS differ from HUMANS across settings and linguistic metrics. However, an important question remains: *when* do LLMs succeed as SIMULATORS of human conversations? In our final experiment, we answer RQ3 by investigating which factors directly impact the differences between SIMULATORS and HUMANS.

### 7.1 Experimental Setup

For this experiment, we focus on depth as opposed to breadth and generate Turn 3 utterances for a random sample of 100,000 conversations from the English-language WildChat corpus. As in §6, we generate responses using the best-performing English prompts from §5 with MISTRAL-LARGE-123B, LLAMA3.1-70B, and MIXTRAL-8x7B. After generating responses, we perform a regression analysis to identify factors linked with higher or lower similarity between SIMULATOR and HUMAN responses. For regression, we use the average similarity across the measures in Table 1 as the dependent variable. For independent variables, we use the conversation metadata (the CHATBOT model used, the region where the HUMAN is located, the SIMULATOR model used, and the prompt), a subset of scalar metrics run on HUMAN Turn 1 and CHATBOT Turn 2 utterances (capitalization, utterance length, politeness, sentiment, subjectivity, toxicity, typographical errors, and average word length), and

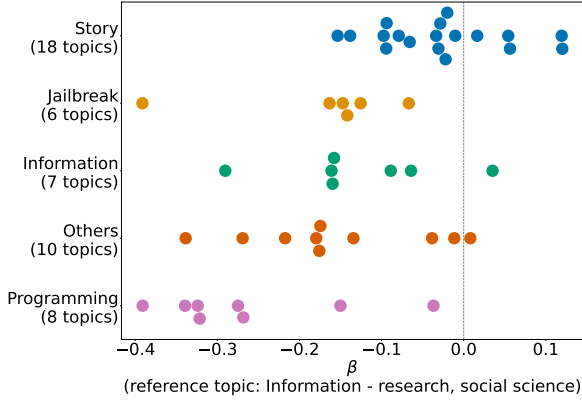


Figure 7: The topic of HUMAN Turn 1 on SIMULATOR influences its performance. Topics are obtained from LDA and manually grouped. We plot the  $\beta$  coefficient of the topic in the regression described in Figure 6.

a set of 50 topics generated using Latent Dirichlet Analysis (LDA) on HUMAN Turn 1 and CHATBOT Turn 2 utterances. Further details on the regression are given in Appendix D.

## 7.2 Results

We first observe that HUMAN Turn 1 has a stronger influence than CHATBOT Turn 2 on SIMULATOR Turn 3 (Figure 6). In other words, the difference between HUMAN Turn 3 and SIMULATOR Turn 3 is explained more by HUMAN Turn 1 than CHATBOT Turn 2. This is likely because two utterances by the same actor (in this case, HUMAN Turns 1 and 3) tend to be linguistically similar, while SIMULATOR Turn 3 has relatively low variation in linguistic features between different conversations.

Accordingly, we find that the performance of Turn 3 simulation is dependent on the simulated HUMAN producing conversations, which are *already* similar to the responses generated by the particular language model and prompt combination. For example, Figure 6 shows that when HUMAN Turn 1 expresses higher politeness, the Turn 3 simulation is predictably more similar, which is likely because LLM responses are more polite by default. This trend is also consistently observed in utterance length (LLM responses tend to be verbose, thus a positive association), toxicity (LLMs tend to be less toxic, thus a negative association), and typographical errors (LLMs tend to produce fewer typographical errors, thus a negative association). This finding echoes the results from §5, where the underlying language model has little impact on similarity (Figure 2), and the prompt primarily influences only syntactic and lexical similarity (Figure 4). There-

fore, capturing the spectrum of linguistic variation that HUMANS naturally express is an open challenge that may require specialized solutions, e.g., prompt engineering or fine-tuning to match the linguistic properties of a target population.

The conversation topic of the initial post (HUMAN Turn 1) is a strong predictor of whether the SIMULATOR can generate similar content in Turn 3 (Figure 7). SIMULATORS tend to be better at story topics, potentially because when the original request is for a story, the conversation often proceeds as a continuation of the story. This continuation shows consistent style and content throughout the dialogue, which makes simulation easier. In contrast, a conversation about a technical topic, e.g., programming, predicts a lower similarity score. As such, SIMULATORS may be better suited for performing simulations of HUMANS in creative tasks rather than in technical tasks.

Overall, we find that SIMULATORS best mirror HUMANS in a narrow set of contexts suited to their safety training and that the models poorly adapt to the range of human speech styles or topics when attempting to generate similar responses (RQ3).

## 8 Discussion and Conclusion

LLM-based simulation of humans in Human-LLM conversations has substantial potential for many applications, such as automated testing and comparison of new models. However, our study has shown that existing open-weight LLMs fall short of simulating these conversations across several metrics. While LLMs perform better at replicating human responses on a semantic level, they encounter difficulties in accurately mirroring human syntax, style, and conversational dynamics. In particular, all the SIMULATORS tend to continue conversations when a human might naturally end them, highlighting a gap in models’ understanding of conversational intent. Our analyses show that the choice of prompt instructions significantly impacts the quality of simulations, often more so than the choice of the SIMULATOR model. Moreover, we find that LLMs struggle in their performance in Chinese and Russian. Finally, we observe that the LLMs’ effectiveness in simulating human behavior is context-dependent: they perform strongly in dialogues that maintain a consistent style, such as storytelling, and weakly in more structured or technical domains like programming.

## 9 Limitations

Simulating human behavior in human-LLM dialogues is inherently challenging due to its open-ended nature, and our study highlights the diverse directions such interactions can take. While we suggest a broad set of diverse prompts, we did not put most of our effort into optimizing those prompts for any specific metric or predefined goal. Our findings indicate that finding the “right” prompt, rather than the “right” model, holds the greatest potential for improvement. Future research could explore whether prompt optimization, tailored to a specific task or metric, yields better results.

In this paper, we decided to simulate the *third turn* in a human-LLM conversation, tasking the SIMULATOR with generating a response based on a short preceding discussion. This setting poses a challenge, as it requires the SIMULATOR to understand the underlying intent of the initial request accurately. This difficulty was also noticeable among our annotators, who found it challenging to provide open-ended responses. Future research could focus on predicting conversation outcomes using a longer seed conversation, which might better capture the nuances and intent of the interaction.

While measuring the similarity between textual content, we use a broad set of metrics to capture a diverse range of language characteristics. However, this list is not exhaustive and can be further modified. Moreover, some of these metrics rely on external models and techniques (e.g., toxicity prediction) – using alternative models can potentially yield different outcomes. This is most relevant for our multilingual experiment. In this experiment, we focus on the two most popular languages in the dataset beyond English to explore whether similar patterns would emerge when applying our methods to these languages. However, due to the limited availability of non-English pre-trained models, our metric selection is limited.

## 10 Ethical Considerations

We use the WildChat dataset (Zhao et al., 2024) as our main data resource for the research. We made sure to follow their ethical guideline while using the data. Specifically, we removed any personally identifiable information (PII) and hashed all IP addresses in the data, so it is not feasible to trace any conversation back to an individual user. As Zhao et al. (2024) mentioned, all WildChat data undergo internal reviews conducted by the AI2 le-

gal team to ensure compliance with data protection laws and ethical standards. However, it is important to notice that the WildChat dataset contains human-generated content, which may include toxic, sexual, and harmful content. Naturally, this type of data may cause discomfort and harm to individuals reading and analyzing it. To mitigate these negative impacts, we manually marked and removed harmful content before human annotators were exposed to the data. Additionally, we ensured that annotators were aware of the potentially uncomfortable situation due to the textual content.

In this research, we use LLMs to simulate human behavior. Although many studies have shown that their outputs are highly “human-like” (Aher et al., 2023; Bang et al., 2023; Liu et al., 2023; Webb et al., 2023), they are prone to problems like generating harmful and biased content. For example, they are known to exhibit political and gender biases (Hartmann et al., 2023; Liu et al., 2024a; Cao et al., 2023) and fail to represent diverse identity groups or cultures (Wang et al., 2024; Tao et al., 2024; Naous et al., 2024). These bottlenecks hinder LLMs’ ability to faithfully represent diverse human behavior, which researchers should be aware of (Abdurahman et al., 2023).

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